20261001 Quiz

for (MATH261) Discrete Mathematics

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Problem 1. Using mathematical induction, prove that

$$2^n \geq n^2$$

for every $n \in \mathbb{N}$ with $n \geq 4$.

Solution. For $n = 4$,

$$2^4 = 16 = 4^2,$$

so the statement holds in the base case.

Assume that, for some $m \geq 4$,

$$2^m \geq m^2.$$

Then

$$2^{m+1} = 2 \cdot 2^m \geq 2m^2.$$

Since $m \geq 4$,

$$2m^2 - (m+1)^2 = m^2 - 2m - 1 = (m-1)^2 - 2 > 0.$$

Hence

$$2^{m+1} \geq 2m^2 > (m+1)^2.$$

Therefore, by mathematical induction,

$$2^n \geq n^2$$

for every $n \in \mathbb{N}$ with $n \geq 4$.

□

Problem 2. Let $n, k \in \mathbb{N}_0$ with $n \geq k$. Prove that

$$\sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

Solution 1. Count the $(k+1)$-element subsets of $[n+1]$ according to their largest element. If the largest element is $i+1$, where $k \leq i \leq n$, then the other k elements must be chosen from $[i]$, giving

$$\binom{i}{k}$$

choices. These cases are disjoint and exhaustive. Therefore,

$$\sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

□

Solution 2. Fix $k \in \mathbb{N}_0$ and use induction on $n \geq k$. For $n = k$,

$$\sum_{i=k}^{k} \binom{i}{k} = \binom{k}{k} = 1 = \binom{k+1}{k+1}.$$

Assume that, for some $m \geq k$,

$$\sum_{i=k}^{m} \binom{i}{k} = \binom{m+1}{k+1}.$$

Then, by Pascal's identity,

$$\begin{aligned}\sum_{i=k}^{m+1} \binom{i}{k} &= \sum_{i=k}^{m} \binom{i}{k} + \binom{m+1}{k} \\ &= \binom{m+1}{k+1} + \binom{m+1}{k} \\ &= \binom{m+2}{k+1}.\end{aligned}$$

Hence the identity holds for every $n \geq k$.

□

Remark. This identity is commonly called the *hockey-stick identity*, since the corresponding entries in Pascal's triangle form a hockey-stick pattern.

Problem 3. Show that every postage amount of at least 12 cents can be formed using only 3-cent and 7-cent stamps.

Solution 1. Let $n \geq 12$. We distinguish the three possible residues of n modulo 3.

(1) If $n \equiv 0 \pmod 3$, then n can be formed using only 3-cent stamps.

(2) If $n \equiv 1 \pmod 3$, then $n \geq 13$ and

$$n = 7 + (n - 7),$$

where $n - 7$ is a nonnegative multiple of 3.

(3) If $n \equiv 2 \pmod 3$, then $n \geq 14$ and

$$n = 14 + (n - 14),$$

where $n - 14$ is a nonnegative multiple of 3.

Thus every $n \geq 12$ can be formed from 3-cent and 7-cent stamps.

□

Solution 2. Use generalized induction. The first three cases are

$$12 = 3 + 3 + 3 + 3, \qquad 13 = 3 + 3 + 7, \qquad 14 = 7 + 7.$$

Now let $m \geq 14$ and assume that the postage amounts $m - 2$, $m - 1$, and m can all be formed. In particular, $m - 2$ can be formed, so adding one 3-cent stamp gives

$$m + 1 = (m - 2) + 3.$$

Hence $m + 1$ can also be formed. Therefore every postage amount of at least 12 cents can be formed.

□

Remark. This is a special case of the *Frobenius coin problem.* For relatively prime integers $a, b > 1$, the largest nonnegative integer not representable as

$$ax + by, \qquad x, y \in \mathbb{N}_0,$$

is called the Frobenius number. In the two-denomination case, Sylvester's formula gives

$$g(a, b) = ab - a - b.$$

Thus $g(3, 7) = 11$, which explains the threshold 12 in this problem.

Problem 4. Let $d \in \mathbb{N}$. Define a relation R on $\mathbb{Z}$ by

$$xRy \quad \iff \quad d \mid (x - y).$$

Show that R is an equivalence relation and determine its equivalence classes.

Solution. For every $x \in \mathbb{Z}$,

$$d \mid (x - x) = 0,$$

so R is reflexive.

If xRy, then $d \mid (x - y)$. Since $y - x = -(x - y)$, we have $d \mid (y - x)$, so yRx. Thus R is symmetric.

If xRy and yRz, then $d \mid (x - y)$ and $d \mid (y - z)$. Hence

$$d \mid ((x - y) + (y - z)) = x - z,$$

so xRz. Thus R is transitive, and therefore R is an equivalence relation.

For $a \in \mathbb{Z}$,

$$\begin{aligned}[a]_R &= \{x \in \mathbb{Z} \mid d \mid (x - a)\} \\ &= \{a + qd \mid q \in \mathbb{Z}\}.\end{aligned}$$

Thus the distinct equivalence classes are

$$[0]_R, [1]_R, \ldots, [d - 1]_R.$$

These are precisely the residue classes modulo d, and xRy is the congruence relation $x \equiv y \pmod{d}$.

□

Problem 5. Prove the following identities. We use the convention $\binom{a}{b} = 0$ when $a, b \in \mathbb{N}_0$ and $b > a$.

(a) For $n, m \in \mathbb{N}_0$ with $n \geq m$,

$$\sum_{k=0}^{m} \binom{n-k}{m-k} = \binom{n+1}{m}.$$

(b) For $u, v, m \in \mathbb{N}_0$,

$$\sum_{k=0}^{m} \binom{u}{k}\binom{v}{m-k} = \binom{u+v}{m}.$$

(c) For $n, m \in \mathbb{N}_0$ with $n \geq m$,

$$\sum_{k=m}^{n} \binom{k}{m}\binom{n}{k} = \binom{n}{m}2^{n-m}.$$

(d) For $n \in \mathbb{N}$ and $0 \leq m \leq n$,

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m}.$$

Solution 1.

(a) Count the m-element subsets of $[n+1]$ according to the first element of $[m+1]$ that is omitted. If $k+1$ is the first omitted element, then $1, \ldots, k$ are selected, $k+1$ is not selected, and the remaining $m-k$ elements are chosen from $\{k+2, \ldots, n+1\}$, which has $n-k$ elements. Hence this case contributes $\binom{n-k}{m-k}$. Since an m-element subset cannot contain all of $[m+1]$, these cases are disjoint and exhaustive. Therefore

$$\sum_{k=0}^{m} \binom{n-k}{m-k} = \binom{n+1}{m}.$$

(b) Let U and V be disjoint sets with $|U| = u$ and $|V| = v$. Count the m-element subsets of $U \cup V$ according to the number k of selected elements from U. For a fixed k, there are

$$\binom{u}{k}\binom{v}{m-k}$$

choices. Summing over k gives the left-hand side, while choosing the m elements directly from $U \cup V$ gives the right-hand side.

(c) Count the pairs (M, K) satisfying

$$M \subseteq K \subseteq [n], \qquad |M| = m.$$

If $|K| = k$, first choose K and then choose $M \subseteq K$, giving $\binom{n}{k}\binom{k}{m}$ choices. Summing over $k = m, \ldots, n$ gives the left-hand side. Alternatively, first choose M in $\binom{n}{m}$ ways. Each of the remaining $n-m$ elements may independently be included in K or not, giving 2^{n-m} choices for K. Thus

$$\sum_{k=m}^{n} \binom{k}{m}\binom{n}{k} = \binom{n}{m}2^{n-m}.$$

(d) Consider all subsets $A \subseteq [n]$ with $|A| \leq m$, assigning weight $(-1)^{|A|}$ to A. Their total weight is

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k}.$$

Pair every $A \subseteq [n-1]$ with $|A| \leq m-1$ with $A \cup \{n\}$. The two members of each pair have opposite weights and cancel. The only unpaired subsets are the m-element subsets of $[n-1]$, of which there are $\binom{n-1}{m}$, each with weight $(-1)^m$. Hence

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m}.$$

This pairing is a sign-reversing involution.

□

Solution 2.

(a) By symmetry of binomial coefficients,

$$\binom{n-k}{m-k} = \binom{n-k}{n-m}.$$

Setting $i = n - k$ and applying Problem 2 gives

$$\begin{aligned}\sum_{k=0}^{m} \binom{n-k}{m-k} &= \sum_{i=n-m}^{n} \binom{i}{n-m} \\ &= \binom{n+1}{n-m+1} \\ &= \binom{n+1}{m}.\end{aligned}$$

(b) By the binomial theorem,

$$(1+x)^u = \sum_{k=0}^{u} \binom{u}{k} x^k, \qquad (1+x)^v = \sum_{j=0}^{v} \binom{v}{j} x^j.$$

Therefore the coefficient of x^m in their product is

$$\sum_{k=0}^{m} \binom{u}{k}\binom{v}{m-k}.$$

Since $(1+x)^u(1+x)^v = (1+x)^{u+v}$, the same coefficient is $\binom{u+v}{m}$.

(c) We have

$$\binom{k}{m}\binom{n}{k} = \binom{n}{m}\binom{n-m}{k-m}.$$

Hence, with $j = k - m$,

$$\begin{aligned}\sum_{k=m}^{n} \binom{k}{m}\binom{n}{k} &= \binom{n}{m} \sum_{j=0}^{n-m} \binom{n-m}{j} \\ &= \binom{n}{m} 2^{n-m},\end{aligned}$$

by the binomial theorem.

(d) For $k \geq 1$, apply Pascal's identity; the $k = 0$ term is $\binom{n}{0} = \binom{n-1}{0} = 1$. Hence

$$\begin{aligned}\sum_{k=0}^{m} (-1)^k \binom{n}{k} &= \sum_{k=0}^{m} (-1)^k \binom{n-1}{k} + \sum_{k=1}^{m} (-1)^k \binom{n-1}{k-1} \\ &= \sum_{k=0}^{m} (-1)^k \binom{n-1}{k} - \sum_{k=0}^{m-1} (-1)^k \binom{n-1}{k} \\ &= (-1)^m \binom{n-1}{m}.\end{aligned}$$

□

Remark. Part (b) is commonly known as *Vandermonde's identity*, or *Vandermonde's convolution*. The same counting argument extends to any finite number of disjoint blocks; for example,

$$\sum_{k_1+\cdots+k_r=m} \binom{n_1}{k_1} \cdots \binom{n_r}{k_r} = \binom{n_1+\cdots+n_r}{m},$$

with the same zero convention for out-of-range binomial coefficients.